Inferring Initial Conditions in the 3-Body Problem

Simulation-Based Inference (SBI) with BayesFlow — Project Plan
TU Dortmund University • Final Project (Topic 7) • Lecturer: Aayush Mishra
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1. Goal and Research Questions

The goal is **not** to solve the three-body problem analytically. It is to use Simulation-Based Inference (estimating unknown parameters from simulated data without an explicit likelihood function) with BayesFlow to infer the **posterior distribution** (what values are plausible after observing data) over the unknown **initial conditions** of a chaotic three-body system, given a noisy observation of its trajectory.

Primary research question

How accurately can SBI recover the initial conditions of a chaotic three-body system, and is the recovered posterior well calibrated?

Secondary research question

How do posterior uncertainty and shape (width, possible multimodality) change with the degree of chaos and with the length of the observed trajectory relative to the Lyapunov time?

2. What the Source Material Tells Us

- **Official spec (Topic 7):** the binding requirements — planar 2D, Newtonian gravity, center-of-mass frame, zero total momentum, fixed distinct masses (not inferred), infer only initial positions and velocities, DOP853 solver with rtol=1e-10 / atol=1e-12, gravitational softening, reject collisions/escapes, observe the trajectory at multiple times, independent Gaussian noise, NPE with jointly trained summary + inference networks.
- **Heggie, *Chaos in the N-body Problem*:** energy conservation is the standard trustworthiness check for a gravitational integrator; chaos is quantified by a positive Lyapunov exponent (exponential divergence rate of nearby orbits).
- **blbadger, *The Three-Body Problem*:** two-body motion is periodic and non-chaotic, but three-body motion is aperiodic and exponentially sensitive to initial conditions, with a stretch-and-fold (Smale-horseshoe) structure — exactly the mechanism that can make our posterior multimodal (several initial conditions folded onto similar trajectories).

3. Precise Requirements Pinned by the Spec

- **Nondimensionalize:** G = 1, total mass M = m1+m2+m3 = 1, characteristic length = 1. With a 1:2:3 ratio the masses are m = (1/6, 2/6, 3/6) = (0.167, 0.333, 0.5).
- **Reference frame:** center of mass at the origin, total momentum zero.
- **Lyapunov timescale is a required step:** empirically estimate the divergence rate λ by evolving two nearby initial conditions, then place several observation times within a few / λ — where the information lives. This is both a spec requirement and the natural knob for the secondary experiment.
- **Noise:** independent Gaussian, applied i.i.d. per recorded timestep, with separate scales σ_x (position) and σ_v (velocity) because they carry different units.
- **Simulator check:** energy conservation over the integration is the mandated trustworthiness test.

4. Key Refinement: Calibration Is the Success Metric

The spec states plainly that *"a wide, possibly multimodal, but well-calibrated posterior is the correct scientific result, not a failure."* So our deliverable is not a narrow posterior — it is an **honestly calibrated** one (when it claims a 90% credible interval, the truth falls inside 90% of the time). Our scientific story is how the width and shape of that honest posterior change with chaos and observation length. Both “chaos widens it” and “more data contracts it” can appear in the same analysis — that is a result, not a contradiction.

5. Timeline (drives our priorities)

- **Slides due:** 19 July 2026 — ~11 days from now.
- **Presentations:** 20–24 July 2026.
- **Report due:** 23 August 2026.
- **Consequence:** top priority is a thin end-to-end pipeline that runs and produces rough calibration/recovery plots quickly, even on a small dataset. We scale up and run the chaos experiment afterward. We do not perfect the simulator before touching BayesFlow.

6. Modeling Decision: Infer the 8 Independent Degrees of Freedom

In 2D there are nominally 12 numbers (3 bodies × 2 position + 2 velocity), but the COM and zero-momentum constraints remove 4, leaving **8 free degrees of freedom**. We infer the positions and velocities of bodies 1 and 2 (8 numbers) and construct body 3 from the constraints:

$$r_3 = -(m_1 \cdot r_1 + m_2 \cdot r_2) / m_3 \qquad v_3 = -(m_1 \cdot v_1 + m_2 \cdot v_2) / m_3$$

Inferring all 12 numbers would create exact degenerate directions that break calibration metrics and confuse the flow. Inferring the 8 free DOF (and reporting all three bodies with body 3 reconstructed) satisfies the spec and keeps the problem well-posed.

7. The Concrete Phased Plan

Phase	What we build	Why / spec link	Report section
0. Scaffold	Isolated Python 3.11/3.12 env, install BayesFlow 2.x + JAX + scipy, repo layout, config.py, requirements.txt	Reproducibility (report requires runnable code)	—
1. Simulator	Softened force law, DOP853 integration, COM + zero-momentum construction, collision/escape rejection, energy-conservation check	Statistical model / how data is generated	Statistical model
2. Lyapunov	Evolve two nearby ICs, measure λ, choose T and observation times t1<...<tK	Spec-mandated; sets where information lives	Statistical model
3. Prior + dataset	Proper priors over the 8 free DOF, noise model (σ_x, σ_v), generate small set then scale to ~20–50k offline to disk	Explicit proper priors; offline regime	Statistical model / Data
4. BayesFlow	Adapter (standardization), summary net (TimeSeriesTransformer/GRU), inference net (CouplingFlow/FlowMatching), ContinuousApproximator, training	NPE with jointly trained networks	Approximator + Training
5. Diagnostics	Loss curves, simulation-based calibration (SBC), posterior contraction, parameter-recovery scatter, posterior predictive checks	Explicitly required diagnostics	Diagnostics
6. Experiment	Observation window relative to Lyapunov time (short/medium/long) + chaos-binned posterior width and calibration	Secondary research question, sharpened	Inference / Results
7. Write-up	Slides (19 Jul) then report (23 Aug)	Deliverables	All

8. Proposed Repository Structure

```
three-body-problem/
  config.py              # masses, G, epsilon, tolerances, T, obs times, noise scales
  src/
    simulator.py         # Phase 1: ICs -> trajectory (+ energy check, rejection)
    lyapunov.py          # Phase 2: estimate lambda, choose observation grid
    priors.py            # Phase 3: sample free DOF, construct full state
    generate_data.py     # Phase 3: build & save offline dataset
    inference.py         # Phase 4: adapter + networks + approximator + train
    diagnostics.py       # Phase 5: SBC, contraction, recovery, PPCs
  experiments/           # Phase 6: observation-length / chaos sweeps
  data/                  # saved simulations (gitignored)
  notebooks/             # exploration + figures for slides/report
  requirements.txt
  README.md
```

9. Immediate Next Step

Phase 0: create an isolated Python 3.11/3.12 environment (the system Python 3.14 is too new for the ML backends), install BayesFlow 2.x + JAX + scipy/numpy/matplotlib, and lay down the repo scaffold + config.py — with a plain-language explanation of why each dependency exists before installing.